

Topics to Research (if you feel like it)

1. Python built-in data structures

In order of importance, look at:

1. Lists
2. Dictionaries
3. Tuples
4. Sets

2. Necessary modules

The only module that I recommend you learn some functions in is `math`. It does math operations, like trigonometric and logarithmic functions. At least learn all your math function equivalents in the `math` module.

3. Time Complexity

Look up Big O, Big Theta, and Big Omega notation. Try to get a high level understanding of it. (MUST understand the basic Python data structures before exploring this)

If you can tell me the big-o notation for indexing a list vs a dictionary or set, you probably understand it well enough.

(`list.sort()` and `sorted(list)` are $n \log n$)

If this goes over your head, that's okay. We'll try to teach you in-club.

4. Helpful modules (not necessary for pyclub, but they'll help you become a better coder)

1. `pathlib` (used to make and manipulate filepaths)
2. `queue` (used to make queues, if you want to be fancier than using a list)
3. `numpy` (it's like `math` but more powerful. It's not in the standard library: you'll have to install it with `pip install numpy`)
4. `threading` (want to run two functions at the same time? Threading is the module for you.)
5. `tkinter` (want to make guis (windowed applications?) `tkinter` is the module for you.)